

Boltzmann Transport equation (BTE) allows us to take a step out of equilibrium thermodynamics. In general the concept of equilibrium implies that there is no net particle flow from one point of a system to another. An equivalent statement is that the electrochemical potential is same throughout the system. Yet, in reality, all electrical devices have currents flowing in them - if it didn't it wouldn't be interesting at all. Our main target is to find expressions for current when "external fields" are present.

In a system of particles, not every particle has the same velocity (momentum). Any system of particles (be it a neutral gas or charged electron gas) is necessarily described by a distribution function. All physically measurable quantities, like electric current, heat current, particle density gradient etc. must involve the correct thermodynamic distribution function. For a system out of equilibrium, this distribution function may change over time or the extent of the sample itself.

What is the equation that governs the change of the distribution function? This question was addressed by Boltzmann. We will make no assumptions (to start with) about whether the distribution function is classical (Maxwell-Boltzmann) or Fermi-Dirac or Bose-Einstein, but try to deduce how it changes due to the presence of an external "force", like an electric field resulting from connecting a battery, or the thermal gradient appearing due to heating one end of the sample. We will see how the presence of electro-magnetic fields, electrochemical potential gradients and thermal gradients drive current. We will do so by trying to calculate the distribution function in a situation slightly away from equilibrium.

To start with we are assuming that the distribution function is a function of $\mathbf{k}$, $\mathbf{r}$, and t , i.e. $f(\mathbf{k}, \mathbf{r}, t)$. The rate of change of $f(\mathbf{k}, \mathbf{r}, t)$, due to changes in any of the variables can then be written as

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \dot{\mathbf{r}} \cdot \nabla_{\mathbf{r}} f + \dot{\mathbf{k}} \cdot \nabla_{\mathbf{k}} f \quad \dots (1)$$

$$\text{or,} \quad \frac{df}{dt} = \frac{\partial f}{\partial t} + \mathbf{v}_{\mathbf{k}} \cdot \nabla_{\mathbf{r}} f + \frac{\mathbf{F}}{\hbar} \cdot \nabla_{\mathbf{k}} f \quad \dots (2)$$

The first term on the right hand side of Eqn. (1) makes sense only if f itself is time-dependent. In Eqn. (2), $\mathbf{v}_{\mathbf{k}}$ refers to the group velocity which is $\mathbf{k}$ -dependent ($\mathbf{v}_{\mathbf{k}} = \frac{1}{\hbar} \nabla_{\mathbf{k}} \epsilon_{\mathbf{k}}$). Note that $\dot{\mathbf{k}} = \frac{\mathbf{F}}{\hbar}$, where $\mathbf{F}$ is the externally applied force. If these terms were the only driving forces for change of f , we would have particles accelerating under applied fields and therefore, time-dependent (non-constant) currents. But, in *steady state*, we get constant current because collisions (scattering) ensures that the change of f due to the external perturbations is exactly balanced by collisions. Let us call the contribution to the rate of change of f as $\left. \frac{df}{dt} \right|_{coll}$. Then in steady state we can write,

$$\frac{\partial f}{\partial t} + \mathbf{v}_{\mathbf{k}} \cdot \nabla_{\mathbf{r}} f + \frac{\mathbf{F}}{\hbar} \cdot \nabla_{\mathbf{k}} f = \left. \frac{df}{dt} \right|_{coll} \quad \dots (3)$$

Relaxation time approximation: To account for the collisions, the simplest model considers

$$\left. \frac{df}{dt} \right|_{coll} = -\frac{\delta f}{\tau}, \text{ where } \delta f = f - f^0 \quad \dots (4)$$

where f^0 is the equilibrium distribution function and τ is the relaxation time. τ will depend on the nature of the scattering but in the above approximation, it is clear from the nature of the collision term that it acts like a restoring force. You see this if you consider the BTE for the case where both $\nabla_r f$ and $\mathbf{F}$ are zero.

Combining all the terms the BTE therefore reads

$$\frac{\partial f}{\partial t} + \mathbf{v}_k \cdot \nabla_r f + \frac{\mathbf{F}}{\hbar} \cdot \nabla_k f = -\frac{\delta f}{\tau} \quad \dots (5)$$

To relate the BTE with electric/magnetic/electromagnetic fields and thermal/conc. gradients let us now consider the Fermi-Dirac distribution, so that

$$f^0 = \frac{1}{1 + e^{\frac{\varepsilon - \mu}{k_B T}}} \quad \dots (6)$$

Here, μ is the electrochemical potential ($\sim$ Fermi Level). Both μ and T are functions of $\mathbf{r}$ (to account for thermal/conc. gradients).

It should be simple to see that

$$\frac{\partial f^0}{\partial \varepsilon_k} = -\beta(1 - f^0)f^0, \text{ where } \beta = (k_B T)^{-1} \quad \dots (7)$$

$$\nabla_r f^0 = \left(\frac{\partial f^0}{\partial \varepsilon_k} \right) \left[-\nabla \mu - (\varepsilon - \mu) \frac{\nabla T}{T} \right] \quad \dots (8)$$

$$\nabla_k f^0 = \left(\frac{\partial f^0}{\partial \varepsilon_k} \right) \nabla_k \varepsilon_k = \hbar \mathbf{v}_k \left(\frac{\partial f^0}{\partial \varepsilon_k} \right) = -\beta f^0(1 - f^0) \hbar \mathbf{v}_k \quad \dots (9)$$

With these preparations, we can now look at the effects of different fields and gradients.

Case 1: When an electric field $\mathbf{E}$ (only) is applied to the system (no thermal/concentration gradients)

$$\mathbf{F} = q\mathbf{E} \quad \dots (10)$$

Therefore, in steady state, Eqn. (5) reads

$$\frac{q\mathbf{E}}{\hbar} \cdot \nabla_k f = -\frac{\delta f}{\tau} \quad \dots (11)$$

Writing $f = f^0 + \delta f$, (11) can be re-written as

$$\frac{q\mathbf{E}}{\hbar} \cdot \nabla_{\mathbf{k}}(f^0 + \delta f) = -\frac{\delta f}{\tau} \quad \dots (12)$$

We make the first order approximation by taking the derivative around the equilibrium value, i.e. set $\frac{q\mathbf{E}}{\hbar} \cdot \nabla_{\mathbf{k}}(\delta f) \approx 0$, so that (12) reads

$$\delta f = -\frac{q\mathbf{E}\tau}{\hbar} \cdot \nabla_{\mathbf{k}} f^0 \quad \dots (13)$$

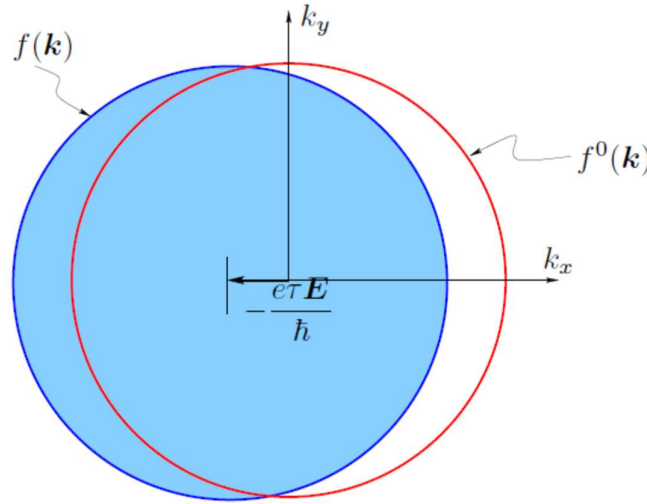
or

$$f = f^0 - \frac{q\mathbf{E}\tau}{\hbar} \cdot \nabla_{\mathbf{k}} f^0 \quad \dots (14)$$

Note that the term $-\frac{q\mathbf{E}\tau}{\hbar} = \delta\mathbf{k} = \dot{\mathbf{k}}\delta t$, if we consider $\delta t = \tau$. Therefore Eqn. (14) suggests just by inspection that,

$$f(\mathbf{k}) = f^0(\mathbf{k} + \delta\mathbf{k}), \quad \text{where } \delta\mathbf{k} = -\frac{q\mathbf{E}\tau}{\hbar}$$

What it essentially means is that if we considered the Fermi sphere centred around $\mathbf{k} = 0$, the effect of the electric field is to shift the origin by $\delta\mathbf{k} = -\frac{q\mathbf{E}\tau}{\hbar}$ (See figure below).



Now we can calculate the current density as,

$$\mathbf{j} = \frac{q}{4\pi^3} \int d^3k \mathbf{v}_{\mathbf{k}} \delta f \quad \dots (15)$$

If we substitute the value of δf from equation (13), we get

$$\mathbf{j} = \frac{q}{4\pi^3} \int d^3k \mathbf{v}_{\mathbf{k}} \left(-\frac{q\mathbf{E}\tau}{\hbar} \cdot \nabla_{\mathbf{k}} f^0 \right) \quad \dots (16)$$

From equation (9)

$$\mathbf{j} = \frac{q}{4\pi^3} \int d^3k \mathbf{v}_{\mathbf{k}} \left(-\frac{q\mathbf{E}\tau}{\hbar} \cdot \hbar \mathbf{v}_{\mathbf{k}} \left(\frac{\partial f^0}{\partial \epsilon_{\mathbf{k}}} \right) \right) \quad \dots (17)$$

or

$$\mathbf{j} = \left\{ \frac{q^2}{4\pi^3} \int \tau d^3k \mathbf{v}_k \otimes \mathbf{v}_k \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) \right\} \cdot \mathbf{E} \quad \dots (18)$$

Here, $A = B \otimes C$ means $A_{ij} = B_i C_j$

Clearly, the expression within the parenthesis is the conductivity tensor, since it relates the current density $\mathbf{j}$ to the applied electric field $\mathbf{E}$, i.e. if we write

$$\mathbf{j} = \vec{\sigma} \mathbf{E} \quad \dots (19)$$

then, we see that

$$\vec{\sigma} = \left\{ \frac{q^2}{4\pi^3} \int d^3k \mathbf{v}_k \otimes \mathbf{v}_k \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) \right\} \quad \dots (20)$$

While equation (20) is the general expression for $\vec{\sigma}$, we can recover the Drude conductivity of metals (modelled as free electron gas) by considering a isotropic system with parabolic ε_k versus k dispersion ($\varepsilon = \hbar^2 k^2 / 2m$). In this case

$$\mathbf{v}_k = \frac{1}{\hbar} \nabla_k \varepsilon_k = \frac{\hbar \mathbf{k}}{m} \quad \dots (21)$$

Therefore

$$\sigma_{ij} = \left\{ \frac{q^2}{4\pi^3} \int \tau \left(\frac{\hbar}{m} \right)^2 k_i k_j \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) d^3k \right\} \quad \dots (22)$$

It is left for you to prove that $\sigma_{ij} = 0 \forall i \neq j$

$$\begin{aligned} \sigma_{ii} &= \left\{ \frac{q^2}{4\pi^3} \int \tau \left(\frac{\hbar}{m} \right)^2 k_i^2 \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) d^3k \right\} \\ &= \left\{ \frac{q^2}{4\pi^3} \int \tau \left(\frac{\hbar}{m} \right)^2 \frac{k_x^2 + k_y^2 + k_z^2}{3} \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) d^3k \right\} \\ &= \left\{ \frac{q^2}{4\pi^3} \int \tau \left(\frac{\hbar}{m} \right)^2 \frac{k^2}{3} \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) d^3k \right\} \quad \dots (23) \end{aligned}$$

Now,

$$\frac{1}{4\pi^3} \int d^3k \equiv \int g(\varepsilon) d\varepsilon$$

where $g(\varepsilon)$ is the density of states. Also $\varepsilon = \hbar^2 k^2 / 2m$.

Therefore

$$\sigma_{ii} = \frac{2q^2}{3m} \int \tau \varepsilon g(\varepsilon) \left(-\frac{\partial f^0}{\partial \varepsilon} \right) d\varepsilon \quad \dots (24)$$

Now, we make the approximation $-\frac{\partial f^0}{\partial \varepsilon_k} = \beta f^0(1 - f^0) = \delta(\varepsilon - \mu)$. This is a valid approximation as $T \rightarrow 0$. At room temperature, this approximation may be done to get the first order term in electrical conductivity (which is the Drude conductivity)

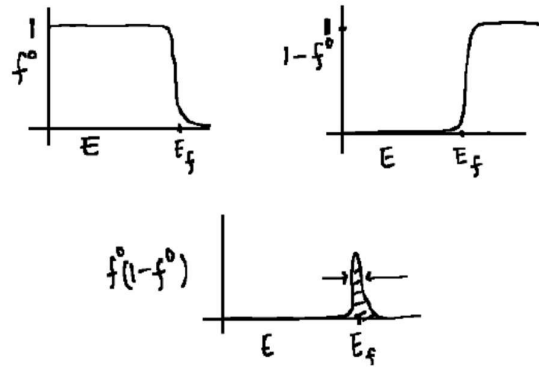


Figure 3.12: How does the product $f^0(1 - f^0)$ behave?

Problem: Show (using the expression of 3D density of states and assuming $\mu(T \sim 300 \text{ K}) = \mu(T = 0) = \varepsilon_F$) that Equation 24 simplifies to

$$\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = \frac{ne^2\tau}{m}$$

So, for a Drude metal (free electron gas)

$$\begin{pmatrix} j_x \\ j_y \\ j_z \end{pmatrix} = \begin{pmatrix} \frac{ne^2\tau}{m} & 0 & 0 \\ 0 & \frac{ne^2\tau}{m} & 0 \\ 0 & 0 & \frac{ne^2\tau}{m} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}$$

This is nothing but Ohm's law.

Case 2: When an electric field E and a magnetic field B are applied to the system (no thermal/concentration gradients)

See next page:

which will add up to zero. So we need to calculate only the diagonal terms. Since there is nothing to distinguish the x, y or z directions, all the diagonal components must be equal. This allows us to write:

$$\begin{aligned}\mu_{ii} &= \frac{q}{n} \int dE D(E) \tau \left(\frac{\hbar}{m} \right)^2 \frac{k_x^2 + k_y^2 + k_z^2}{3} \delta(E - E_f) \\ &= \frac{q}{3n} \int dE D(E) \tau \frac{2E}{m} \delta(E - E_f)\end{aligned}\quad (3.127)$$

Using the expression for density of states in 3D, eqn 3.127 reduces to:

$$\begin{aligned}\mu_{ii} &= \frac{q}{3\pi^2 m n} \int dE k^3 \tau \delta(E - E_f) \\ &= \frac{q\tau}{m} \quad \text{since } k_F^3 = 3\pi^2 n\end{aligned}\quad (3.128)$$

Electric and magnetic field together

Now let's recall eqn 3.110 and allow a magnetic field. Eqn 3.112 that described the deviation of the distribution function from equilibrium should now read :

$$f(\mathbf{k}) = f^0(\mathbf{k}) - \frac{q\tau}{\hbar} (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \nabla_{\mathbf{k}} f^0 \quad (3.129)$$

Now because the force term has explicit dependence on $\mathbf{k}$ we can no longer write down the solution by inspection, as we did in eqn 3.113. However we now try a solution of the same form, with an unknown vector $\mathbf{Z}$. Our target is to write $\mathbf{Z}$ as a function of $\mathbf{E}$ and $\mathbf{B}$, but free of $\mathbf{k}$ and $\mathbf{v}_{\mathbf{g}}$. Thus we want $\mathbf{Z}$, such that

$$f(\mathbf{k}) = f^0(\mathbf{k} - \frac{q\tau}{\hbar} \mathbf{Z}) \quad (3.130)$$

Hence,

$$\delta f = -\frac{q\tau}{\hbar} \mathbf{Z} \cdot \nabla_{\mathbf{k}} f^0 \quad (3.131)$$

We now use the assumed form (eqn 3.131) with eqns 3.109 and 3.110. This gives:

$$\frac{q}{\hbar} (\mathbf{v} \times \mathbf{B}) \cdot (\nabla_{\mathbf{k}} f^0 + \nabla_{\mathbf{k}} \delta f) + \frac{q}{\hbar} \mathbf{E} \cdot \nabla_{\mathbf{k}} f^0 = -\frac{\delta f}{\tau} \quad (3.132)$$

We already know that $\nabla_{\mathbf{k}} f^0$ points along $\mathbf{v}_{\mathbf{g}}$ and hence the first term in eqn 3.132 gives zero. This leaves us with

$$\frac{q}{\hbar} (\mathbf{v} \times \mathbf{B}) \cdot \nabla_{\mathbf{k}} \delta f + \frac{q}{\hbar} \mathbf{E} \cdot \nabla_{\mathbf{k}} f^0 = \frac{q}{\hbar} \mathbf{Z} \cdot \nabla_{\mathbf{k}} f^0 \quad (3.133)$$

Now we need to calculate $\nabla_{\mathbf{k}} \delta f$.

(Please follow the red signs below, wherever applicable)

$$\begin{aligned}\nabla_{\mathbf{k}} \delta f &= -\nabla_{\mathbf{k}} \frac{q\tau}{\hbar} \mathbf{Z} \cdot \nabla_{\mathbf{k}} f^0 \\ &= -\frac{q\tau}{\hbar} \nabla_{\mathbf{k}} (-\beta f^0 (1 - f^0) \mathbf{Z} \cdot \hbar \mathbf{v}_{\mathbf{g}}) \\ &= +\beta q\tau ((1 - f^0)(\mathbf{Z} \cdot \mathbf{v}_{\mathbf{g}}) \nabla_{\mathbf{k}} f^0 + f^0 (\mathbf{Z} \cdot \mathbf{v}_{\mathbf{g}}) \nabla_{\mathbf{k}} (1 - f^0) + f^0 (1 - f^0) \nabla_{\mathbf{k}} (\mathbf{Z} \cdot \mathbf{v}_{\mathbf{g}}))\end{aligned}\quad (3.134)$$

Once again the first two terms in the RHS of 3.134 will give zero when dotted with $\mathbf{v} \times \mathbf{B}$ as they are $\propto \mathbf{v}_{\mathbf{g}}$. The only term left is

$$\nabla_{\mathbf{k}} \mathbf{Z} \cdot \mathbf{v}_{\mathbf{g}} = \nabla_{\mathbf{k}} \mathbf{Z} \cdot \frac{\hbar \mathbf{k} - q \mathbf{A}}{m} = \frac{\hbar}{m} \nabla_{\mathbf{k}} \mathbf{Z} \quad (3.135)$$

In eqn 3.135, $\mathbf{A}$ denotes the vector potential of the magnetic field, $\mathbf{v}_g$ is related to the canonical momentum in presence of a magnetic field in the usual way. Combining eqns 3.134 and 3.135 we can write:

$$(\mathbf{v} \times \mathbf{B}) \cdot \nabla_k \delta f = +\beta q \tau f^0 (1 - f^0) (\mathbf{v} \times \mathbf{B}) \cdot \frac{\hbar}{m} \mathbf{Z} \quad (3.136)$$

So eqn 3.133 now simplifies to:

$$\begin{aligned} & +\frac{\hbar}{m} \beta q \tau f^0 (1 - f^0) (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{Z} + (\mathbf{E} - \mathbf{Z}) \cdot \nabla_k f^0 = 0 \\ \therefore & +\frac{\hbar}{m} \beta q \tau f^0 (1 - f^0) (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{Z} + (\mathbf{E} - \mathbf{Z}) \beta f^0 (1 - f^0) \hbar \mathbf{v}_g = 0 \\ \therefore & \frac{q\tau}{m} (\mathbf{v}_g \times \mathbf{B}) \cdot \mathbf{Z} + (\mathbf{E} - \mathbf{Z}) \cdot \mathbf{v}_g = 0 \\ \therefore & \frac{q\tau}{m} (\mathbf{B} \times \mathbf{Z}) \cdot \mathbf{v}_g + (\mathbf{E} - \mathbf{Z}) \cdot \mathbf{v}_g = 0 \\ \therefore & \mathbf{E} = \mathbf{Z} + \frac{q\tau}{m} \mathbf{B} \times \mathbf{Z} \end{aligned} \quad (3.137)$$

We call $\mathbf{Z}$ as the Hall vector. When both $\mathbf{E}$ and $\mathbf{B}$ fields are present, this quantity in some way, "replaces" the electric field in the transport equation. But we still need to express $\mathbf{Z}$ explicitly in terms of $\mathbf{E}$ and $\mathbf{B}$, with $\mu = q\tau/m$. The proof is left as an exercise.

$$\mathbf{Z} = \frac{\mathbf{E} + \mu \mathbf{B} \times \mathbf{E} + \mu^2 (\mathbf{B} \cdot \mathbf{E}) \mathbf{B}}{1 + \mu^2 B^2} \quad (3.138)$$

PROBLEM: If $\mathbf{E} = \mathbf{Z} - \mathbf{A} \times \mathbf{Z}$, then show that

$$\mathbf{Z} = \frac{\mathbf{E} + \mathbf{A} \times \mathbf{E} + (\mathbf{A} \cdot \mathbf{E}) \mathbf{A}}{1 + A^2}$$

Hint : Try $\mathbf{A} \times \mathbf{E}$ and $\mathbf{A} \cdot \mathbf{E}$

In the Hall geometry, $\mathbf{B} = (0, 0, B)$ and $\mathbf{E} = (E_x, E_y, 0)$. To calculate the (magneto) conductivity tensor in the Hall geometry now we can make use of the relation

$$\mathbf{j} = \vec{\sigma} \mathbf{Z}$$

It is left as a homework to show (making use of Eqn. 3.138) that,

$$\begin{pmatrix} j_x \\ j_y \\ j_z \end{pmatrix} = \frac{\sigma_0}{1 + (\omega_c \tau)^2} \begin{pmatrix} 1 & -\omega_c \tau & 0 \\ \omega_c \tau & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ 0 \end{pmatrix}$$

where $\sigma_0 = \frac{ne^2\tau}{m}$.

Case 3: When no electric field E or magnetic field B are applied to the system but there are thermal and or concentration gradients

1) Recall the BTE (Equation 5):

$$\frac{\partial f}{\partial t} + \mathbf{v}_k \nabla_r f + \frac{\mathbf{F}}{\hbar} \nabla_k f = -\frac{\delta f}{\tau} \quad \dots (25)$$

Since, $\mathbf{E} = \mathbf{B} = 0$, the BTE reads (in steady state)

$$\mathbf{v}_k \nabla_r f = -\frac{\delta f}{\tau} \quad \dots (26)$$

or

$$\mathbf{v}_k \nabla_r f^0 = -\frac{\delta f}{\tau} \quad \dots (27)$$

Problem: Verify that for the Fermi-Dirac distribution (Equation 8)

$$\nabla_r f^0 = \nabla_r f^0 = \left(\frac{\partial f^0}{\partial \varepsilon_k} \right) \left[-\nabla \mu - (\varepsilon_k - \mu) \frac{\nabla T}{T} \right] \quad \dots (28)$$

Therefore,

$$\delta f = \tau \mathbf{v}_k \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) \left[-\nabla \mu - (\varepsilon_k - \mu) \frac{\nabla T}{T} \right]$$

Now, we can calculate the current density as (Eqn. 15)

$$\mathbf{j} = \frac{q}{4\pi^3} \int d^3k \mathbf{v}_k \delta f \quad \dots (29)$$

or

$$\mathbf{j} = \left\{ \frac{q}{4\pi^3} \int \tau d^3k \mathbf{v}_k \otimes \mathbf{v}_k \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) \right\} \cdot \left[-\nabla \mu - (\varepsilon_k - \mu) \frac{\nabla T}{T} \right] \quad \dots (30)$$

or

$$\begin{aligned} \mathbf{j} = & \left\{ \frac{q^2}{4\pi^3} \int \tau d^3k \mathbf{v}_k \otimes \mathbf{v}_k \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) \right\} \cdot \left(-\frac{\nabla \mu}{q} \right) \\ & + \left\{ \frac{q^2}{4\pi^3} \int \tau d^3k \mathbf{v}_k \otimes \mathbf{v}_k \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) (\varepsilon_k - \mu) \right\} \cdot \left(-\frac{\nabla T}{qT} \right) \quad (31) \end{aligned}$$

Note that the expression in the curly brackets of the first term in RHS is the same as $\vec{\sigma}$ (Equation 20). It relates the charge current to the gradient of the chemical potential.

Note: If we also had an external electric field applied to the system, i.e. $\mathbf{E} \neq 0$, Eqn. 31 still reads the same, **and not**

$$\mathbf{j} = \left\{ \frac{q^2}{4\pi^3} \int \tau d^3k \mathbf{v}_k \otimes \mathbf{v}_k \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) \right\} \cdot \left(\mathbf{E} - \frac{\nabla \mu}{q} \right) + \left\{ \frac{q^2}{4\pi^3} \int \tau d^3k \mathbf{v}_k \otimes \mathbf{v}_k \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) (\varepsilon_k - \mu) \right\} \cdot \left(-\frac{\nabla T}{qT} \right) \quad (32)$$

This needs to be understood carefully. If we started with the situation $\mathbf{E} \neq 0$ and $\nabla_r f \neq 0$, we would have had to consider (instead of Eqn. 28)

$$\nabla_r f^0 = \nabla_r f^0 = \left(\frac{\partial f^0}{\partial \varepsilon_k(r)} \right) \left[\nabla(\varepsilon_k(r) - \mu(r)) - (\varepsilon_k(r) - \mu(r)) \frac{\nabla T}{T} \right] \dots (33)$$

In other words, $\mathbf{E} \neq 0$ (be it internal or external fields), would result in the spatial dependence of ε (band bending). So, one would get (as the first term of Eqn. 32) as

$$\mathbf{j} = \left\{ \frac{q^2}{4\pi^3} \int \tau d^3k \mathbf{v}_k \otimes \mathbf{v}_k \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) \right\} \left(\mathbf{E} - \frac{\nabla \mu}{q} + \frac{\nabla \varepsilon_k(r)}{q} \right) \dots (34)$$

However,

$$\frac{\nabla \varepsilon_k(r)}{q} = \nabla \phi(r) = -\mathbf{E} \quad \dots (35)$$

Thus, Eqn. 30 is the right expression of current even in presence of externally applied electric fields. What is to be understood is that no matter what the source of the electric field is, it will be reflected in an equivalent band bending. When both these terms are considered, they would cancel each other out. In still other words, the band bending happens in response to an electric field (internal or external). We can neglect both contributions due to the electric field and the band bending and consider the current to be given by the gradient of the chemical potential only.

When we discuss p - n junctions in equilibrium, we will see that there is no current, even though there is an electric field (due to the internal built-in potential) and also band bending. Important is to note that $\nabla \mu = 0$, i.e. the Fermi level remains constant across the junction at equilibrium.

Equation (31) can be written as

$$\mathbf{j} = \vec{\sigma} \left(-\frac{\nabla \mu}{q} \right) + \left\{ \frac{q^2}{4\pi^3} \int \tau d^3k \mathbf{v}_k \otimes \mathbf{v}_k \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) (\varepsilon_k - \mu) \right\} \left(-\frac{\nabla T}{qT} \right) \quad (36)$$

or more generally

$$\mathbf{j} = \vec{\mathbf{L}}_{11} \left(-\frac{\nabla\mu}{q} \right) + \vec{\mathbf{L}}_{12} (-\nabla T) \quad \dots (37)$$

Here, $\vec{\mathbf{L}}_{11}$ and $\vec{\mathbf{L}}_{12}$ are known as the **transport coefficients**, since they relate *charge* current to electric and thermal fields, respectively. Later, we will see how to relate *heat* current ($\mathbf{j}_Q$) to the same fields, in terms of the two other transport coefficients $\vec{\mathbf{L}}_{21}$ and $\vec{\mathbf{L}}_{22}$. The transport equations in general can thus be written as

$$\begin{Bmatrix} \mathbf{j} \\ \mathbf{j}_Q \end{Bmatrix} = \begin{bmatrix} \vec{\mathbf{L}}_{11} & \vec{\mathbf{L}}_{12} \\ \vec{\mathbf{L}}_{21} & \vec{\mathbf{L}}_{22} \end{bmatrix} \begin{Bmatrix} -\frac{\nabla\mu}{q} \\ -\nabla T \end{Bmatrix} \quad \dots (38)$$

At this stage, let us define

$$\vec{\mathbf{l}}^{(\alpha)} = \left\{ \frac{q^2}{4\pi^3} \int \tau d^3k \mathbf{v}_k \otimes \mathbf{v}_k \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) (\varepsilon_k - \mu)^\alpha \right\} \quad \dots (39)$$

We see that

$$(1) \quad \vec{\mathbf{L}}_{11} = \vec{\sigma} = \vec{\mathbf{l}}^{(0)} \quad \dots (40)$$

$$(2) \quad \vec{\mathbf{L}}_{12} = \frac{\vec{\mathbf{l}}^{(1)}}{qT} \quad \dots (41)$$

If we are interested in the transport coefficients of a free electron gas (isotropic ε_k versus k dispersion), it suffices to calculate $\vec{\mathbf{l}}^{(\alpha)}$. In case of the free electron gas, only the diagonal elements of $\vec{\mathbf{l}}^{(\alpha)}$ are non-zero (and equal), i.e.

$$l_{xx}^{(\alpha)} = l_{yy}^{(\alpha)} = l_{zz}^{(\alpha)} = \left\{ \frac{q^2}{4\pi^3} \int \tau d^3k \left(\frac{\hbar}{m} \right)^2 \frac{k^2}{3} \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) (\varepsilon_k - \mu)^\alpha \right\} \quad \dots (42)$$

or

$$l_{xx}^{(\alpha)} = l_{yy}^{(\alpha)} = l_{zz}^{(\alpha)} = \left\{ \frac{2q^2}{3m} \int \tau \varepsilon_k g(\varepsilon_k) \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) (\varepsilon_k - \mu)^\alpha d\varepsilon_k \right\} \quad \dots (43)$$

Clearly, for $\alpha = 0$, we recover the expression for σ as in Eqn. 24. However, now we will not make the approximation $-\frac{\partial f^0}{\partial \varepsilon_k} = \beta f^0(1 - f^0) = \delta(\varepsilon_k - \mu)$, which is strictly valid as $T \rightarrow 0$. Instead we will evaluate $l^{(0)}$ by substituting the expression for $g(\varepsilon_k)$ for the free electron gas

$$g(\varepsilon_k) = \frac{\sqrt{2}m^{3/2}}{\hbar^3\pi^2} \sqrt{\varepsilon_k} \quad \dots (44)$$

So that (43) can be written as (for $\alpha = 0$)

$$l^{(0)} = \xi \int_0^\infty \varepsilon_k \sqrt{\varepsilon_k} \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) d\varepsilon_k \quad \dots (45)$$

Here

$$\xi = \frac{2q^2}{3m} \cdot \frac{\sqrt{2}m^{3/2}}{\hbar^3 \pi^2} \cdot \tau \quad \dots (46)$$

The problem then simplifies to solving the integrals of the type

$$\int_0^\infty \varepsilon_k^s \left(\frac{\partial f^0}{\partial \varepsilon_k} \right) d\varepsilon_k \quad \dots (47)$$

Integrating by parts, we get

$$\int_0^\infty \varepsilon_k^s \left(\frac{\partial f^0}{\partial \varepsilon_k} \right) d\varepsilon_k = \varepsilon_k^s f^0 \Big|_0^\infty - s \int_0^\infty \varepsilon_k^{s-1} f^0 d\varepsilon_k = -s \int_0^\infty \varepsilon_k^{s-1} f^0 d\varepsilon_k \quad \dots (48)$$

The first term is dropped since $f^0(\varepsilon_k \rightarrow \infty) = 0$.

Therefore,

$$\int_0^\infty \varepsilon_k^s \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) d\varepsilon_k = s \int_0^\infty \varepsilon_k^{s-1} f^0 d\varepsilon_k \quad \dots (49)$$

By defining reduced chemical potential $\tilde{\mu} = \beta\mu$ and $\tilde{\varepsilon}_k = \beta\varepsilon_k$, we can write

$$\int_0^\infty \varepsilon_k^s f^0 d\varepsilon_k = \left(\frac{1}{\beta} \right)^{s+1} F_s(\tilde{\mu}) \quad \dots (50)$$

The integrals $F_s(\tilde{\mu}) = \int_0^\infty \tilde{\varepsilon}_k^s f^0(\tilde{\varepsilon}_k, \tilde{\mu}) d\tilde{\varepsilon}_k$ are called the **Fermi (Dirac) Integrals**.

In case of metals, $\tilde{\mu} \gg 1$. So, one may expand $F_s(\tilde{\mu})$ as

$$F_s(\tilde{\mu}) = \frac{\tilde{\mu}^{s+1}}{s+1} + s\tilde{\mu}^{s-1} \frac{\pi^2}{6} + \dots \quad (51)$$

Problem: Calculate σ without the approximation $-\frac{\partial f^0}{\partial \varepsilon_k} = \beta f^0(1 - f^0) = \delta(\varepsilon_k - \mu)$ but making use of Eqn. (51) and show that the first-order term in $\tilde{\mu}$ yields the same value as with the above approximation.

$$\text{Ans: } \sigma = \xi \mu^{3/2} \left[1 + \frac{\pi^2}{8} \left(\frac{k_B T}{\mu} \right)^2 \right]$$

Seebeck Coefficient calculation: Seebeck effect refers to the generation of a (open circuit) voltage in response to a temperature gradient. The Seebeck co-efficient is the ratio of the open circuit voltage to the temperature gradient.

Setting $\mathbf{j} = 0$ and $\mathbf{j}_Q = 0$ in Eqn. 37 and identifying $-\frac{\nabla\mu}{q} = \mathbf{E}$ we get the ratio of $\mathbf{E}$ and ∇T to be

$$\vec{S} = \vec{L}_{12} \vec{L}_{11}^{-1} \quad \dots (52)$$

This is the Seebeck co-efficient.

Problem: Show that the Seebeck Coefficient S for the free electron gas can be written as

$$S = \frac{1}{qT} \left[\frac{5}{3} k_B T \frac{F_{3/2}(\tilde{\mu})}{F_{1/2}(\tilde{\mu})} - \mu \right]$$

For $\tilde{\mu} \gg 1$, show that

$$S = \frac{\pi^2}{3} \frac{k_B T}{\mu} \left(\frac{k_B}{q} \right)$$

Thermal current and Thermal Conductivity:

The *heat* transport equation is given by (Eqn. 38)

$$\mathbf{j}_Q = \vec{L}_{21} \left(-\frac{\nabla\mu}{q} \right) + \vec{L}_{22} (-\nabla T) \quad \dots (53)$$

To relate $\vec{L}_{21}$ and $\vec{L}_{22}$ to $\vec{L}^{(\alpha)} = \left\{ \frac{q^2}{4\pi^3} \int \tau d^3k \mathbf{v}_k \otimes \mathbf{v}_k \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) (\varepsilon_k - \mu)^\alpha \right\}$, we need to find an expression for $\mathbf{j}_Q$ in terms of quantities we can calculate in the BTE formalism: This is obtained from the thermodynamic relation

$$dQ = TdS \text{ (At a fixed T)}$$

But

$$TdS = dU - \mu dN$$

So

$$dQ = dU - \mu dN$$

Therefore,

$$\dot{Q} = \dot{U} - \mu \dot{N}$$

or

$$\mathbf{j}_Q = \mathbf{j}_u - \mu \mathbf{j}_n$$

Thus the heat current may be related to the (internal) energy current and the number current. Therefore

$$\mathbf{j}_Q = \frac{1}{4\pi^3} \int d^3k \mathbf{v}_k \varepsilon_k \delta f - \mu \frac{1}{4\pi^3} \int d^3k \mathbf{v}_k \delta f \quad \dots (54)$$

We now substitute the value of

$$\delta f = \tau \mathbf{v}_k \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) \left[-\nabla \mu - (\varepsilon_k - \mu) \frac{\nabla T}{T} \right]$$

and replace $\frac{1}{4\pi^3} d^3k$ with $g(\varepsilon) d\varepsilon$ to get the expression of $\mathbf{j}_Q$ as [\(verify this\)](#)

$$\begin{aligned} \mathbf{j}_Q = & \frac{1}{q} \left\{ \frac{q^2}{4\pi^3} \int \tau d^3k \mathbf{v}_k \otimes \mathbf{v}_k \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) (\varepsilon_k - \mu) \right\} \cdot \left(-\frac{\nabla \mu}{q} \right) \\ & + \frac{1}{q^2 T} \left\{ \frac{q^2}{4\pi^3} \int \tau d^3k \mathbf{v}_k \otimes \mathbf{v}_k \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) (\varepsilon_k - \mu)^2 \right\} \cdot (-\nabla T) \end{aligned} \quad \dots (55)$$

We can thus identify,

$$\vec{L}_{22} = \frac{\vec{l}^{(2)}}{q^2 T}$$

and

$$\vec{L}_{21} = \frac{\vec{l}^{(1)}}{q}$$

To calculate the thermal conductivity we notice that it is the coefficient which relates $\mathbf{j}_Q$ to $(-\nabla T)$, for $\mathbf{j} = 0$

$\mathbf{j} = 0$ implies

$$-\frac{\nabla \mu}{q} = \vec{L}_{12} \vec{L}_{11}^{-1} (\nabla T) = \vec{S} (\nabla T) \quad \dots (56)$$

and therefore, the thermal conductivity is

$$\vec{K} = (\vec{L}_{22} - \vec{L}_{21} \vec{S}) \quad \dots (57)$$

Since $\vec{S}$ is small for metals, $\vec{K}$ can be equated to $\vec{L}_{22}$.

For metals, K can be calculated by evaluating $L_{22} = \frac{l^{(2)}}{q^2 T}$ (or to be precise $(L_{22} - L_{21} S)$).

However, here we will relate K to the electronic specific heat defined as

$$C_{el} = \int g(\varepsilon_k) \left(\frac{\partial f^0}{\partial T} \right) (\varepsilon_k - \mu) d\varepsilon_k \quad \dots (58)$$

which will allow us to recover the Drude-Sommerfeld expression for K as

$$K = \frac{1}{3} C_{el} l_F v_F \quad \dots (59)$$

Here v_F is the Fermi velocity and $l_F = v_F \tau$ is the mean free path at the Fermi energy.

Note that

$$L_{22} = \frac{l^{(2)}}{q^2 T} = \frac{1}{q^2 T} \left\{ \frac{2q^2}{3m} \int \tau \varepsilon_k g(\varepsilon_k) \left(-\frac{\partial f^0}{\partial \varepsilon_k} \right) (\varepsilon_k - \mu)^2 d\varepsilon_k \right\} \quad \dots (60)$$

Verify that

$$\frac{\partial f^0}{\partial \varepsilon_k} = -\frac{T}{(\varepsilon_k - \mu)} \frac{\partial f^0}{\partial T} \quad (61)$$

When we replace $-\frac{\partial f^0}{\partial \varepsilon_k}$ with $\frac{T}{(\varepsilon_k - \mu)} \frac{\partial f^0}{\partial T}$ in equation (60), we get

$$L_{22} = \left\{ \frac{2}{3m} \int \tau \varepsilon_k g(\varepsilon_k) \left(\frac{\partial f^0}{\partial T} \right) (\varepsilon_k - \mu) d\varepsilon_k \right\} \quad (62)$$

Since the electrons in a narrow energy band at the Fermi level $\mu = \varepsilon_F$ take part in the thermal conduction, it is not a bad approximation to replace ε_k in the integrand with ε_F and assume τ to be independent of ε_k . We then get

$$L_{22} = \frac{2\tau \varepsilon_F}{3m} \int g(\varepsilon_k) \left(\frac{\partial f^0}{\partial T} \right) (\varepsilon_k - \mu) d\varepsilon_k \quad \dots (63)$$

i.e.

$$L_{22} = \frac{2\tau}{3m} \varepsilon_F C_{el} = \frac{2\tau}{3m} \cdot \frac{1}{2} m v_F^2 C_{el} = \frac{1}{3} C_{el} l_F v_F \quad (64)$$

This calculation reveals that the Drude-Sommerfeld result is correct if the small Seebeck coefficient of metals is ignored and if electrons only at ε_F are assumed to contribute to the thermal conductivity. However, the second statement cannot be (strictly speaking) true as for those electrons $(\varepsilon_k - \mu)$ would be zero!